F FWallSim · Firewall Course

CLASS 9

Identity-based access with LDAP

LDAP server, user groups, identity-based policy

Intermediate | ~25 minutes | FWallSim v1.1.0 (FortiOS 7.4 style)

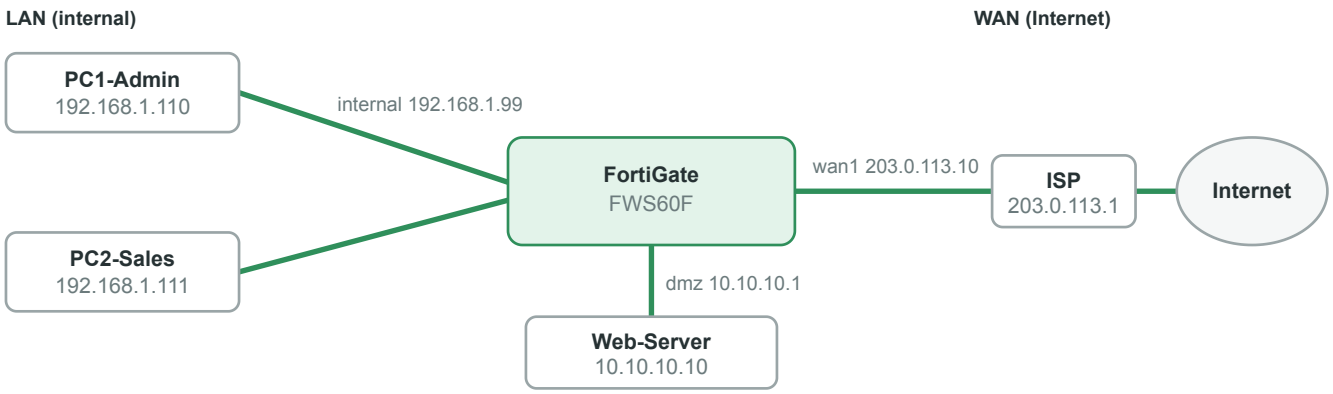
YOU WILL LEARN

- LDAP server
- user groups
- identity-based policy

BEFORE YOU START

Finish Class 8 first.

The lab network



The goal

THE IDEA

Only the Sales team may use the internet. Connect the FortiGate to Active Directory (AD-Server 10.10.10.20) and make the internet policy require the Sales group. The directory details are on User & Authentication > LDAP Servers.

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **Identity-based access with LDAP** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **An LDAP server object connects to AD successfully**

   User & Authentication > LDAP Servers > Create New, then Test Connectivity until it says Successful.

2. **A user group contains the LDAP server, limited to CN=Sales**

   User Groups > Create New: Members = your LDAP server, Remote Groups: CN=Sales,OU=Groups,DC=lab,DC=local.

3. **Policy 1 requires that group**

   Edit policy 1 > Source: add the user group next to LAN-Subnet.

4. **jsmith logs in on PC1 and can browse**

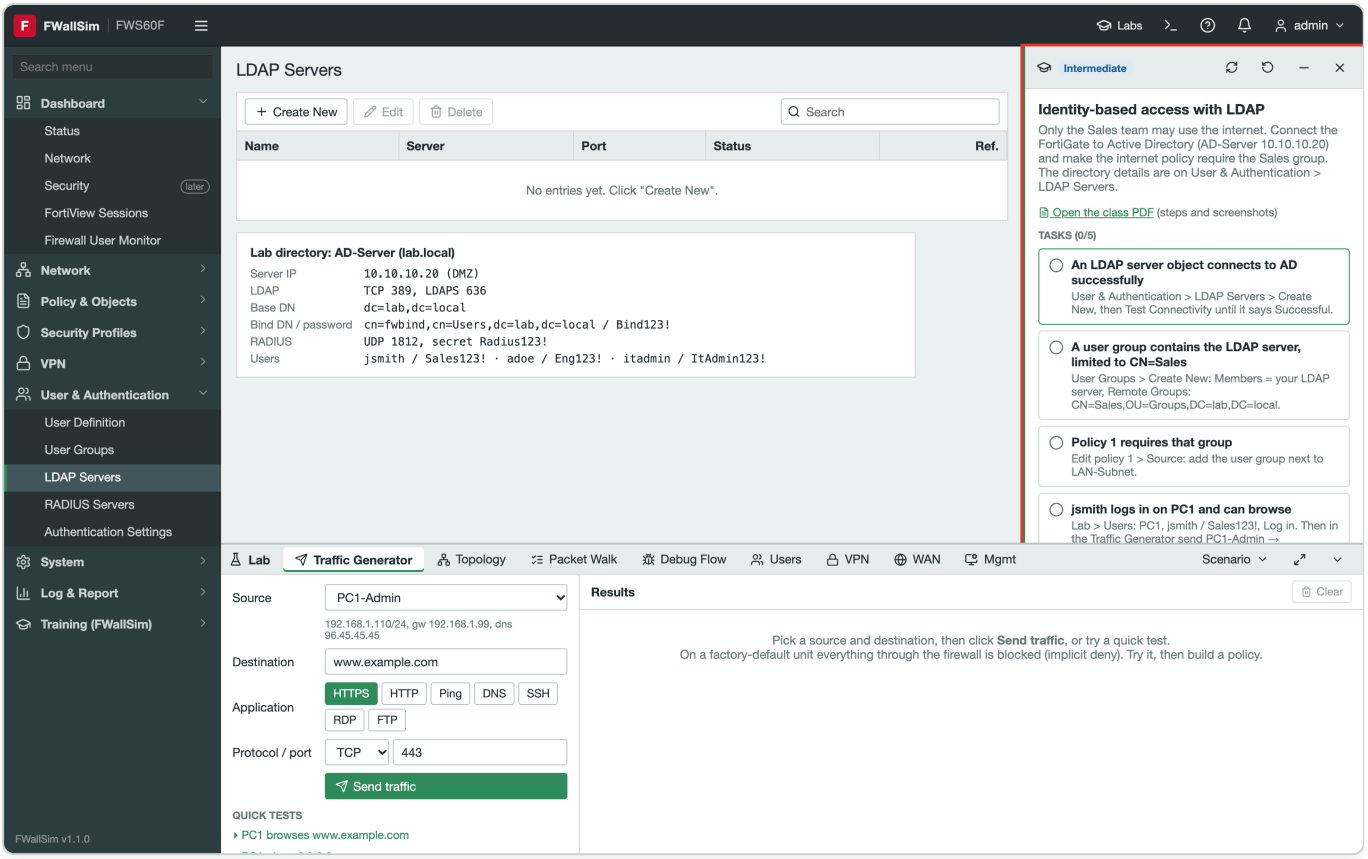
   Lab > Users: PC1, jsmith / Sales123!, Log in. Then in the Traffic Generator send PC1-Admin → www.example.com (HTTPS). Note: before login even DNS is refused, because DNS cannot show a login page. In production, add a DNS-only policy without groups above this one.

5. **adoe (Engineering) logs in on PC2 and is refused**

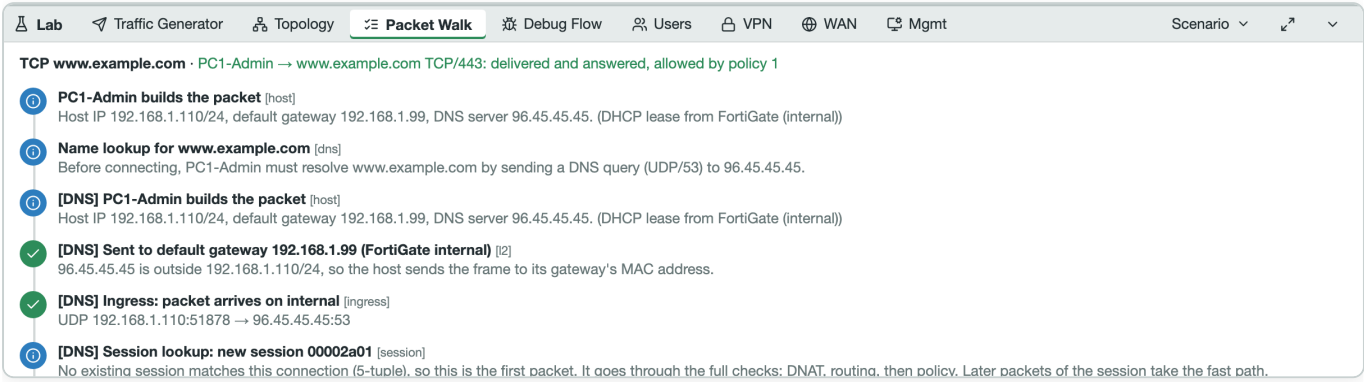
   Lab > Users: PC2, adoe / Eng123!, Log in. Then in the Traffic Generator send PC2-Sales → www.example.com (HTTPS). It must be refused.

Screenshots

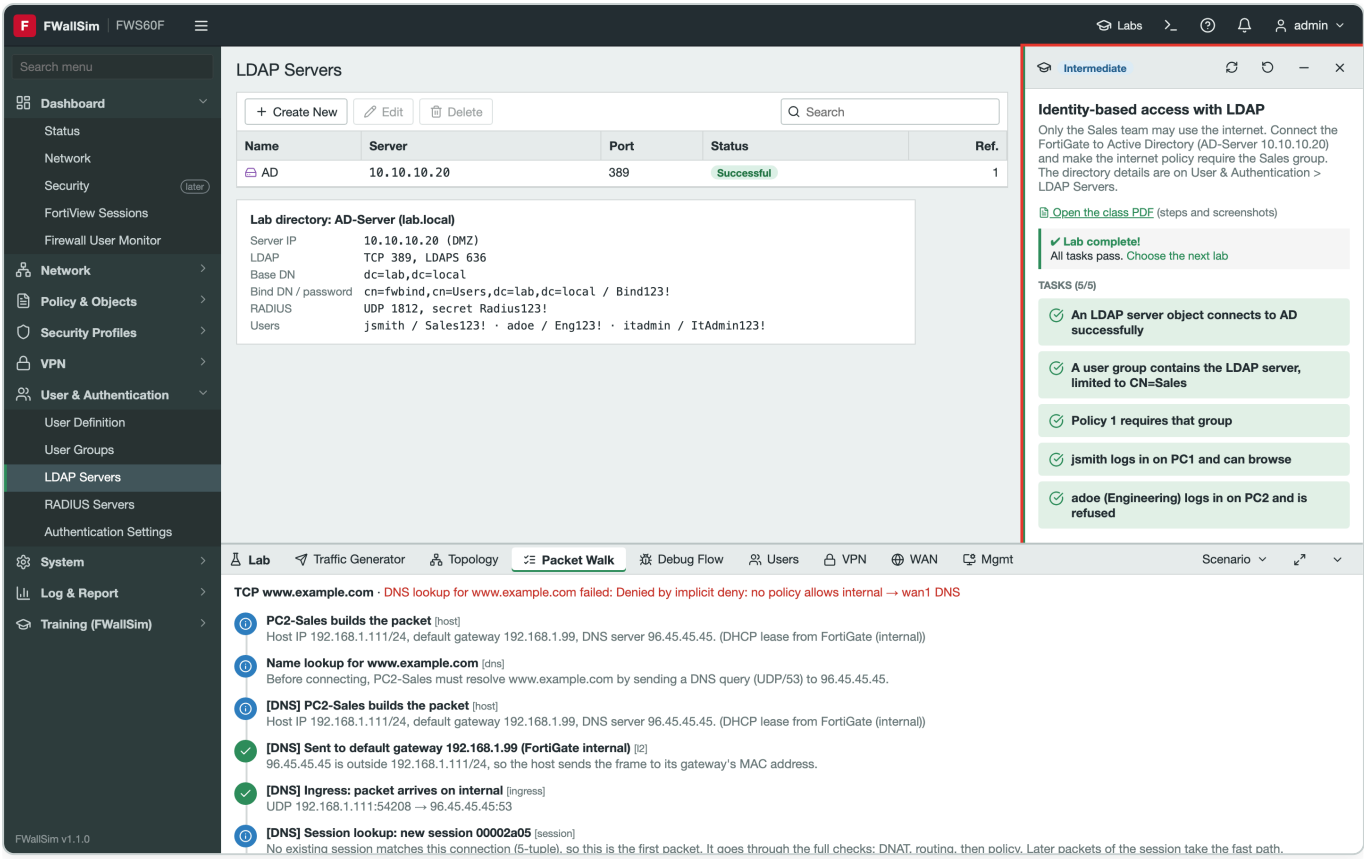
Numbered red markers show where to click or what to look at.



Lab Started



Packet Walk Before



Lab Solved

LDAP Servers

+ Create New | Edit | Delete | Search

Name	Server	Port	Status	Ref.
AD	10.10.10.20	389	Successful	1

Lab directory: AD-Server (lab.local)

Server IP	10.10.10.20 (DMZ)
LDAP	TCP 389, LDAPS 636
Base DN	dc=lab,dc=local
Bind DN / password	cn=fwbind,cn=Users,dc=lab,dc=local / Bind123!
RADIUS	UDP 1812, secret Radius123!
Users	jsmith / Sales123! · adoe / Eng123! · itadmin / ItAdmin123!

LDAP Servers

User Groups

+ Create New | Edit | Delete | Search

Group Name	Group Type	Members	Remote Groups	Ref.
Guest-group	Firewall			0
Sales	Firewall	AD	AD: CN=Sales,OU=Groups,DC=lab,DC=local	1

Firewall policies match **groups**, not individual logins. Members can be local users, remote users, or a whole LDAP/RADIUS server. A "Remote Group" limits a server member to users in that directory group.

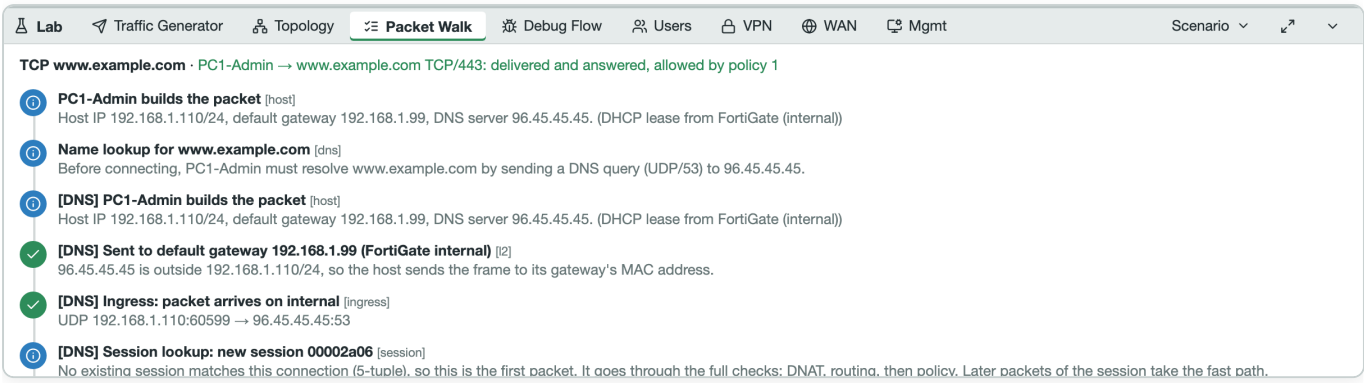
User Groups

Firewall User Monitor

User Name	User Group	IP Address	Device	Method	Logged in	Expires in	
jsmith	Sales	192.168.1.110	PC1-Admin	LDAP (AD)	2026/11/13 19:19:31	298 s	Deauthenticate
adoe	(none)	192.168.1.111	PC2-Sales	LDAP (AD)	2026/11/13 19:19:31	298 s	Deauthenticate

The FortiGate remembers a login per **source IP**. While an IP is listed here, its traffic matches identity-based policies for the user's groups. CLI: `diagnose firewall auth list`.

Firewall User Monitor



Packet Walk After

Next: Class 10: Site-to-site IPsec VPN